

Module 3

Version Control Systems and Collaboration

Objectives

At the end of the chapter, the learner should be able to:

- Understand the purpose of version control in software projects.
- Use Git to create repositories, commit changes, and track project history.
- Explain how version control facilitates team collaboration.

What is version control?

- Version control, also known as source control, is the practice of tracking and managing changes to software code.
- Version control systems are software tools that help software teams manage changes to source code over time.

What is Git?

- By far, the most widely used modern version control system in the world today is Git. Git is a mature, actively maintained open source project originally developed in 2005 by Linus Torvalds, the famous creator of the Linux operating system kernel.



What is Git?



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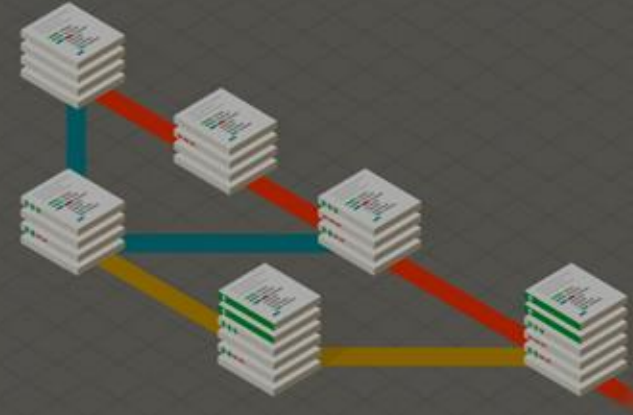
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Git is a **free and open source** distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is **easy to learn** and has a **tiny footprint with lightning fast performance**. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like **cheap local branching**, convenient **staging areas**, and **multiple workflows**.



About

The advantages of Git compared to other source control systems.



Documentation

Command reference pages, Pro Git book content, videos and other material.



Downloads

GUI clients and binary releases for all major platforms.



Community

Get involved! Bug reporting, mailing list, chat, development and more.



Windows GUIs



Tarballs



Mac Build



Source Code



Pro Git by Scott Chacon and Ben Straub is available to **read online for free**. Dead tree versions are available on **Amazon.com**.



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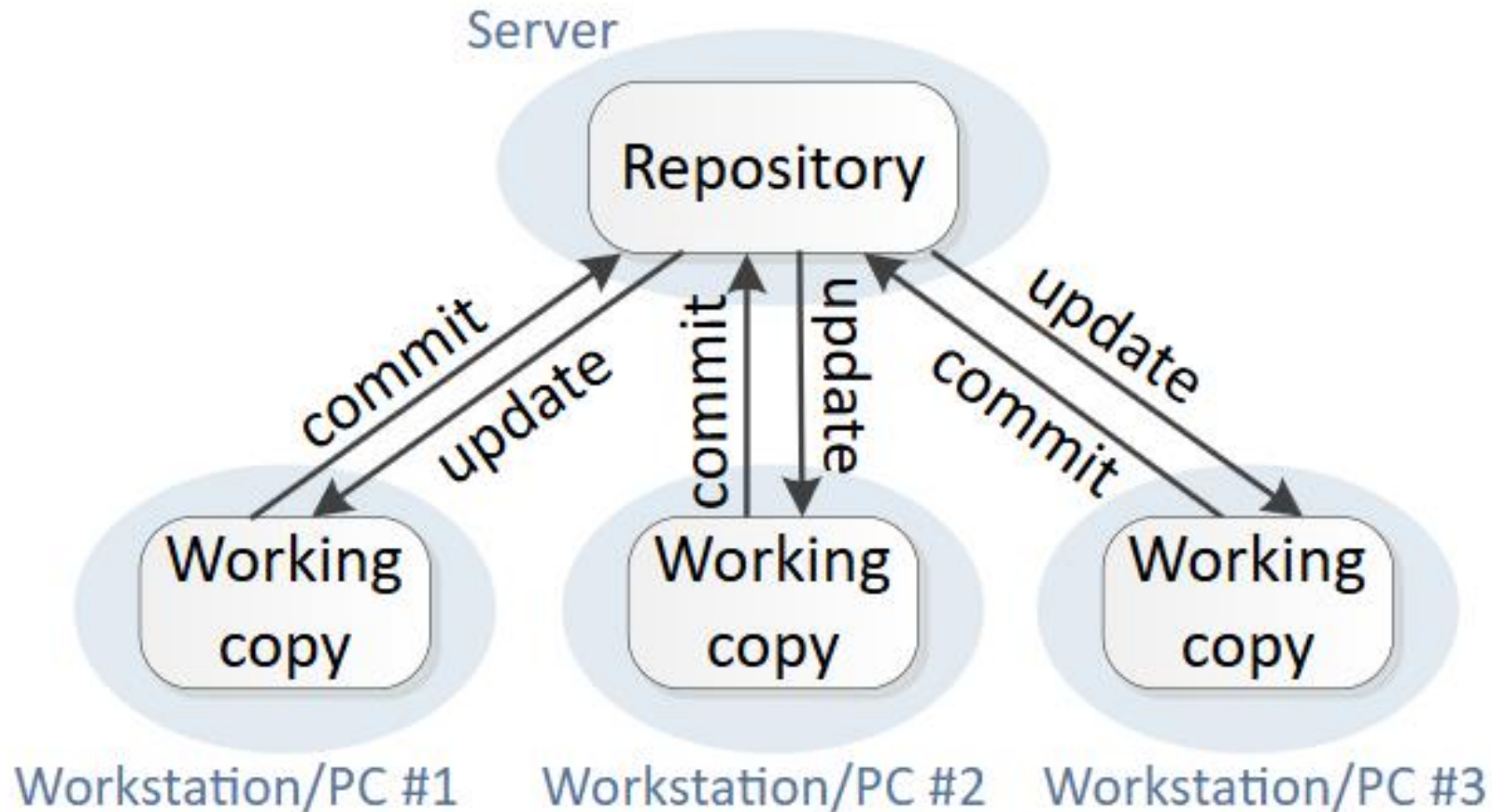
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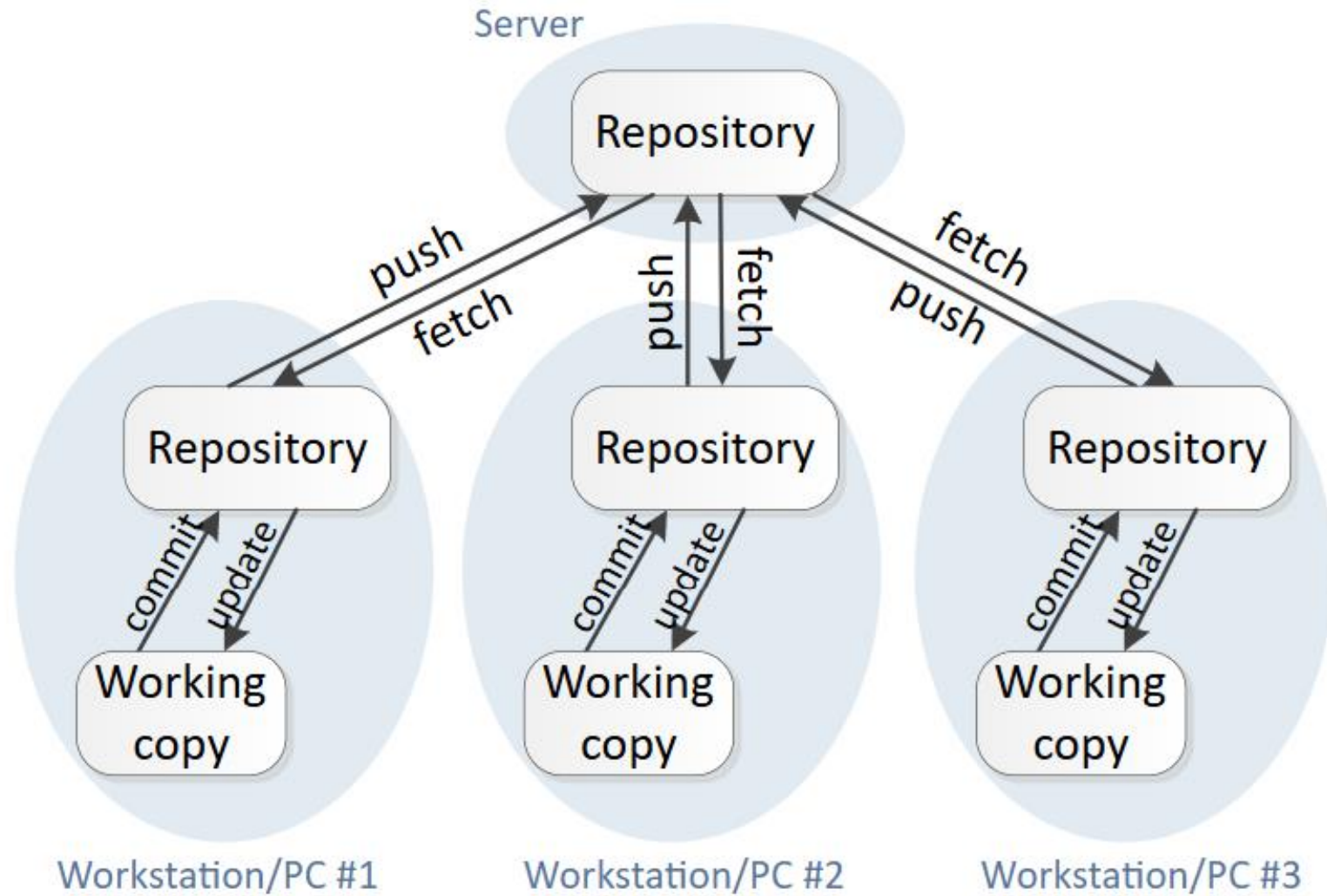
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Centralized version control



Distributed version control in git



Git hosting services



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GIT

VS

GITHUB





Git is a software.

GitHub is a service.





Git is a version control system to manage source code history.

GitHub is a hosting service for Git repositories.





Git is maintained by linux.

GitHub is maintained by
Microsoft.





Git is installed locally
on the system

GitHub is hosted on
the web





Git has no user management feature.

GitHub has a built-in user management feature.





Git is open-source
licensed.

GitHub includes a free-
tier and pay-for-use tier.





Git has minimal external tool configuration.

GitHub has an active marketplace for tool integration.



Git Commands

```
git init
```

initialize an existing directory as a Git repository



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Git Commands

```
git clone [URL]
```

retrieve an entire repository from a hosted location via URL



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Git Commands

```
git clone [URL]
```

retrieve an entire repository from a hosted location via URL



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Git Commands

git status

show modified files in working directory, staged for your next commit



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Git Commands

git status

show modified files in working directory, staged for your next commit



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Git Commands

```
git add [file]
```

add a file as it looks now to your next commit
(stage)



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Git Commands

```
git reset [file]
```

unstage a file while retaining the changes in working directory

Git Commands

```
git diff
```

diff of what is changed but not staged



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Git Commands

```
git diff --staged
```

diff of what is staged but not yet committed



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Git Commands

```
git commit -m “[descriptive message]”
```

commit your staged content as a new commit snapshot



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Visit for more git commands <https://education.github.com/git-cheat-sheet-education.pdf>

GitHub

GIT CHEAT SHEET

Git is the free and open source distributed version control system that's responsible for everything GitHub related that happens locally on your computer. This cheat sheet features the most important and commonly used Git commands for easy reference.

INSTALLATION & GUIs

With platform specific installers for Git, GitHub also provides the ease of staying up-to-date with the latest releases of the command line tool while providing a graphical user interface for day-to-day interaction, review, and repository synchronization.

GitHub for Windows
<https://windows.github.com>

GitHub for Mac
<https://mac.github.com>

For Linux and Solaris platforms, the latest release is available on the official Git web site.

Git for All Platforms
<http://git-scm.com>

STAGE & SNAPSHOT

Working with snapshots and the Git staging area

git status
show modified files in working directory, staged for your next commit
git add [file]
add a file as it looks now to your next commit (stage)
git reset [file]
unstage a file while retaining the changes in working directory
git diff
diff of what is changed but not staged
git diff --staged
diff of what is staged but not yet committed
git commit -m "[descriptive message]"
commit your staged content as a new commit snapshot

SETUP

Configuring user information used across all local repositories

```
git config --global user.name "[firstname lastname]"
```

set a name that is identifiable for credit when review version history

```
git config --global user.email "[valid-email]"
```

set an email address that will be associated with each history marker

```
git config --global color.ui auto
```

BRANCH & MERGE

Isolating work in branches, changing context, and integrating changes

```
git branch
```



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References

- <https://www.atlassian.com/git/tutorials/what-is-version-control>
- <https://www.atlassian.com/git/tutorials/what-is-git>
- <https://homes.cs.washington.edu/~mernst/advice/version-control.html>
- <https://rewind.com/blog/github-vs-bitbucket-vs-gitlab-comparison/>



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